

Markov Chains

Problem 1

The population of Replicant skin cells has a $\frac{4}{5}$ chance of tripling or $\frac{1}{5}$ chance of going extinct. What is the probability Replicant skin cells do not go extinct, given the Replicant population began with only one cell?

Problem 2

Fordo and Frodo are betting opposing wagers of \$1 on unbiased coin tosses, playing until one goes bankrupt. Fordo has \$99 and Frodo has \$66, what are Fordo's chances of losing?

Problem 3

Let's play a game with 2 coins. Coin 1 lands on heads 60% of the time, while coin 2 lands on heads 50% of the time. A coin is continually flipped until it lands on tails, then we switch to the other coin and repeat the coin flips until we have to switch again. If we start the game with coin 1, what is the probability that coin 2 is used on the 5th flip?

Problem 4

Suppose an organisation has n employees, and that each employee has one of three possible jobs, which changes every day according to a Markov chain with the following transition matrix:

$$\begin{pmatrix} 0.7 & 0.2 & 0.1 \\ 0.2 & 0.6 & 0.2 \\ 0.1 & 0.4 & 0.5 \end{pmatrix}.$$

Over the long-term, what percentage of employees will be in each job?

Problem 5

Scientists are measuring the propensity of the Janx species of animals on the planet Anaxes to sleep given the conditions A : sunny, B : sunset/sunrise, C : dark. There is an equal chance of the test beginning with each state. A to B is as likely as C to B , but both are twice as likely as A to A , C to C but A to C is impossible. It is impossible for C to C , C to A , and A to C but B to C , B to A are equally likely on this planet. Now given it is A , there is only a 30% change of falling asleep, 50% for B , and 80% for C .

1. Draw the Hidden Markov Model Diagram
2. What is the overall chance of falling asleep.

Problem 6

**Tintin and Justin play a game of chance where Tintin out of an infinite pile of a_1, a_2, a_3, a_4 cent coins and Justin out of b_1, b_2, b_3, b_4 cent coins randomly chose a denomination each round to add to their pile and whoever first reaches an integer amount wins the match and with it right to buy an All or Nothing Option at \$8 for the next match that pays off \$10 if their pile for that match reaches a winning sum under \$2.

1. Intuitively what is the EV of Tintin and Justin's pile?
2. *If Justin has a greater EV in his pile, how are his odds of profiting compared to Tintin from the All or Nothing Option, assuming he wins the previous match and buys the option?
3. Given Justin has 5, 10, 15, 25 cent coins and Tintin has 1, 2, 5, 10 cent coins, approximate Justin's Expected Profit for the second match, assuming he won the first won and bought the option.
4. What should be a fair price for this option given previous approximations?